ROHITH PAPANA

Electronics & Communication Engineer | Embedded Systems | PCB Design | IoT

✉ papanarohith45@gmail.com | ✆ +91-8309943166

■ Andhra Pradesh, India | GitHub: github.com/Prohith45



OBJECTIVE

Results-driven Electronics & Communication Engineering student with hands-on expertise in embedded systems, PCB design, IoT, and programming. Seeking a role to apply and expand my technical skills while contributing to a forward-thinking organisation.

EDUCATION

B.Tech – Electronics & Communication Engineering — Pursuing
Prasad V. Potluri Siddhartha Institute of Technology (PVPSIT) | JNTUK

Diploma – Electronics & Communication Engineering — 2025
Dhanekula Institute of Engineering and Technology | SBTET — Percentage: 97%

SSC (10th Class) — 2022
SLM Chaithanya High School — Percentage: 84%

TECHNICAL SKILLS

Programming	C, Python
Embedded Platforms	Arduino, Raspberry Pi
Electronics & Hardware	Circuit design & analysis, Assembly & soldering, Troubleshooting, Prototyping
PCB Design	Schematic design, PCB layout & routing, Copper-clad fabrication, Motor drivers & PWM controllers
Tools / Concepts	Flask, gTTS, MIT App Inventor, IoT, CNC machining, SMPS, RF communication

INTERNSHIP EXPERIENCE

Avishkaran Industry Pvt. Ltd., Hyderabad — Intern

Bluetooth Device Manufacturing

- Assembled and tested consumer Bluetooth devices including sound bars, mini speakers, TWS earbuds, and neckbands.
- Worked hands-on with electronic components in a consumer electronics manufacturing environment.
- Applied quality control procedures and learned end-to-end production workflows.

PROJECTS

IoT-Based Audio Announcement System

Diploma Final Year Project

- Built a real-time voice announcement system using Raspberry Pi Zero W2, MIT App Inventor, and Bluetooth speakers.
- Implemented a Flask server with Google Text-to-Speech (gTTS) and Bluetooth audio streaming for multi-room announcements.
- Reduced manual communication effort in institutional environments via mobile-app-driven broadcasts.

Automatic Writing Machine (Handwriting Replication System)

- Designed a CNC-based system using stepper motors, CNC shield, and drivers to replicate personalised handwriting.
- Converted A–Z and 0–9 handwritten samples into a custom font, integrated via a web interface for autonomous writing.

AI-Based Healthcare Assistance Robot

- Designed a smart robot for automated medicine dispensing and patient monitoring based on doctor-prescribed schedules.
- Integrated camera-based medicine identification, movement tracking, obstacle avoidance, and autonomous navigation.

Bench Power Supply (Custom Built)

- Designed and assembled a variable power supply using SMPS and buck converter modules with adjustable voltage/current.
- Integrated a real-time digital display; used for electronic circuit testing and prototyping.

RF-Based Universal Remote (8-Channel)

- Developed an 8-channel wireless remote using HT12E/HT12D encoder–decoder ICs and a 433 MHz RF module.

Custom PCB Design & Fabrication

- Designed and fabricated PCBs on copper-clad boards — H-bridge MOSFET motor drivers and PWM speed controllers.
- Hands-on experience in schematic capture, PCB layout, etching, drilling, and functional testing.

AI Reel Generator

- Built an AI pipeline that generates and publishes short-form video reels (1–2 min) from a single topic prompt.
- Automates script writing, media processing, and content upload for Instagram and YouTube.

LANGUAGES

English	Professional Working Proficiency
Telugu	Native Proficiency

PERSONAL DETAILS

Date of Birth	29 August 2006
Nationality	Indian

HOBBIES & INTERESTS

- Exploring cutting-edge electronics, AI tools, and cybersecurity
- Building and experimenting with innovative electronic gadgets
- Gym & fitness training
- Gaming and outdoor sports

DECLARATION

I hereby declare that all the information furnished above is true, complete, and correct to the best of my knowledge and belief. I understand that any misrepresentation or omission of facts may disqualify me from consideration or result in dismissal.